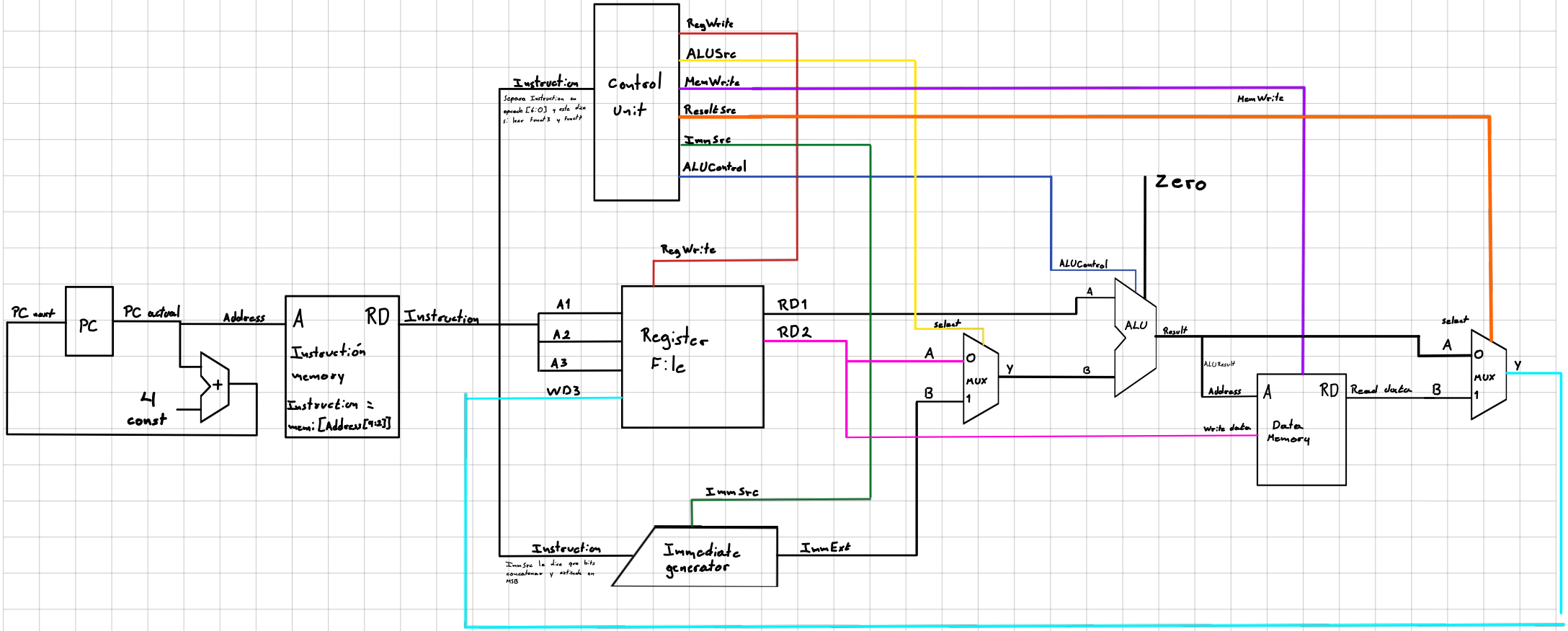

PC next
PC
PC actual
4
const
Address
A
RD
Instruction memory
Instruction = mem[Address[9:2]]
Instruction
Instruction
Separa Instruction en opcode [6:0] y este dice si leer funct3 y funct7
Control Unit
RegWrite
ALUSrc
MemWrite
ResultSrc
ImmSrc
ALUControl
RegWrite
A1
A2
A3
WD3
Register File
RD1
RD2
select
A
B
0
MUX
1
Y
ALUControl
A
ALU
Zero
Result
MemWrite
ALUResult
Address
A
RD
Data Memory
Write data
Read data
select
A
B
0
MUX
1
Y
ImmSrc
Instruction
ImmSrc le dice que bits concatenar y extender en MSB
Immediate generator
ImmExt